

Human Computer Interaction 2 MIDTERMS

TOPIC 5 — DESIGN SYSTEMS AND PROTOTYPING

1. Low-Fidelity Wireframing — **Basic** and **simplified** sketches or layouts that represent the structure of a webpage or app screen without design details.

- Black and white or grayscale
- No colors, images, or detailed typography
- Uses boxes, placeholders, and basic shapes
- Focuses on layout and content hierarchy

Tools: Paper and pencil, Balsamiq, Figma

2. High-Fidelity Mockups — **Detailed** and **polished** designs that closely resemble the final product in terms of visuals and layout.

- Includes colors, fonts, images, and icons
- Reflects branding and design system
- Often used for final approval before development

Tools: Figma, Adobe XD

3. Paper Prototyping — **Hand-drawing interfaces** on paper and using them to simulate user interactions.

Purpose:

- Quick and inexpensive way to test ideas
- Used in early brainstorming stages

How It Works:

- Draw screens on separate sheets
- Use gestures or flip pages to simulate navigation
- Test concepts with users before digital work

4. Digital Wireframing — **Computer-based layouts** of a user interface that show structure and functionality without visual design.

Purpose:

- Bridges low-fidelity sketches and high-fidelity mockups
- Can include basic interactions

Tools: Figma, Adobe XD, Sketch

5. Clickable Prototypes — **Interactive versions** of designs that simulate how users would navigate and interact with the product.

Features:

- Interactive buttons, links, and menus
- Navigation between pages/screens
- Transitions and animations

Used For:

- User testing
- Presenting flows to stakeholders

6. User Flow Creation — **Visualizes the step-by-step path** users take to complete tasks within an app or website.

Components:

- Starting point
- Decision points
- End goals

Purpose:

- Helps map logical screen-to-screen movement
- Ensures a smooth and intuitive experience

7. Interaction Design — **Defines how users engage with elements** in the UI, such as buttons, menus, and transitions.

Focus Areas:

- Feedback, such as hover effects and button states
- Micro-interactions, such as loading indicators and animations
- Usability and responsiveness

Goals:

- Enhance user satisfaction and efficiency
- Make interfaces intuitive and engaging

Tools: Figma, Principle, ProtoPie

TOPIC 6 — CREATING REUSABLE COMPONENTS

1. Reusable Components — Design elements such as buttons, cards, or navbars that **can be used multiple times** across a project.

Purpose:

- Ensure consistency
- Save time
- Allow updates to be applied to all instances

2. Component Variants — Multiple versions of a component **grouped into one organized set**.

Examples: Default, hover, disabled, different button sizes or states.

3. Component Properties — Controls that **allow changes** to text, colors, or visibility **without detaching** the component.

4. Component Libraries — Collections of reusable components that **can be accessed across multiple files** or projects.

Purpose:

- Promote standardization
- Improve consistency
- Support team collaboration

5. Design Systems — A collection of reusable components, styles, and guidelines that define the visual language and standards of a product.

Purpose: Ensure consistency and efficiency in design and development.

6. Style Guides — Documents that describe the visual and functional elements of a design.

Includes:

- Typography
- Colors
- UI elements

Purpose: Serve as a reference for designers and developers to maintain consistency.

7. Naming Conventions — Consistent rules for naming and organizing components, layers, and variables.

Purpose:

- Easier searching
- Better organization
- Easier management
- Improved collaboration

Key Practices:

- Use forward slashes to group components
- Use descriptive names for variables
- Use consistent casing such as Title Case or camelCase for properties

TOPIC 7 — FIGMA FEATURES

1. Auto Layout — A Figma feature that allows frames to adapt to their content by automatically adjusting size or spacing between items.

Purpose: Create responsive and flexible designs.

2. Constraints — Rules that define how an object resizes or moves when its parent frame is resized.

Key Features:

- Can pin objects to the top, bottom, left, or right
- Can scale objects proportionally

Purpose: Create flexible and responsive layouts.

3. Vertical and Horizontal Scroll — Features used to create prototypes that simulate scrolling like a real website or app.

Overflow Behavior:

- Vertical
- Horizontal
- Both

Uses:

- Simulate long pages
- Create horizontal carousels
- Simulate infinite scrolling
- Simulate horizontal swiping galleries

4. Variables — Reusable, named values that can be applied to design properties.

Examples: Colors, numbers, strings, and booleans.

Purpose: Foundation of a scalable design system.

5. Modes — Different sets of variables that can be applied to a design.

Purpose: Quickly switch between themes or configurations.

Examples: Light mode, dark mode, different content languages.

6. Component Interactions — Connect components and frames to simulate user actions.

Examples: Clicks, hovers, and drags.

Purpose:

- Build realistic clickable prototypes
- Create complex user flows
- Test concepts
- Share interactive prototypes for feedback

7. Overlays — Frames that appear on top of the main content.

Examples: Dropdown menu, tooltip.

Purpose: Display temporary information or controls.

8. Modals — A specific type of overlay that usually requires the user's attention or action before continuing.

Examples: Sign-up form, confirmation dialog.

Purpose: Focus the user's attention on specific content or action.

9. Boolean Operations — Used to combine or cut vector shapes to create new and more complex shapes.

Types:

Union — Combines two or more shapes into a single shape.

Subtract — Removes one shape from another.

Intersect — Keeps only the overlapping area of two or more shapes.

Exclude — Keeps everything except the overlapping area of two or more shapes.

10. Figma Plugins — Third-party extensions that add new functionality to Figma.

Uses:

- Automate tasks
- Import external content
- Enhance workflow

TOPIC 8 — LAYOUT AND RESPONSIVE DESIGN

1. Grid-Based Layout
2. Flexible and Responsive Design
3. Mobile-First Approach
4. Breakpoints and Screen Size
5. Touch Interface Design
6. Cross-Platform Considerations
7. Device-Specific Guidelines